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Research Paper

Optic Nerve Edema in Pediatric Middle Cranial Fossa Arachnoid Cysts: Report of 51 Patients From a Single Institution

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ABSTRACT

Background: Middle fossa arachnoid cysts (MFACs) are rare, congenital lesions that may rupture and cause symptoms of elevated intracranial pressure. We sought to describe the presence of and factors associated with optic nerve edema in MFACs, focusing on the utility of ophthalmologic evaluations for guiding cyst management.
Methods: We reviewed clinical and radiographic information for all patients with MFACs with ophthalmologic evaluations at our institution. Headache, cranial nerve palsy, emesis, altered mental status, fatigue, and seizures were considered MFAC-related symptoms. Univariate and multivariable analyses evaluated factors associated with optic edema.
Results: Fifty-one patients between 2003 and 2022 were included. Cysts were a median volume of 169.9 cm^3 (interquartile range: 70.5, 647.7). Evidence of rupture with subdural hematoma/hygroma occurred in 19 (37.3%) patients. Eighteen (35.3%) patients underwent surgery for their cyst and/or rupture-associated intracranial bleed. Eleven (21.6%) patients had optic edema; all were symptomatic and experienced cyst rupture. Ten of these patients received surgery. Postoperatively, optic edema resolved in 80% of cases. Cyst volume and symptoms were not associated with optic edema; however, patients with ruptured cysts, particularly those with traumatic rupture, were more likely to have optic edema and receive surgery ($P < 0.001$).
Conclusions: We found optic edema in 21.6% of evaluated MFACs, and this comprised of 57.9% of ruptured cases. Optic edema was not found in unruptured cysts. Cyst fenestration improved optic edema and patient symptoms. In conjunction with clinical history and neuroimaging, optic edema may help guide MFAC management, particularly in patients with cyst rupture.

Introduction

Arachnoid cysts are rare, congenital lesions that are often incidentally discovered on neuroimaging in the absence of symptoms.[1-4] Approximately 2.6% of all arachnoid cysts occur in children, and although these cysts are not limited to any one intracranial location, up to 50% can develop in the middle cranial fossa.[1,2,5-7] In spite of their usually benign nature, on occasion, pediatric middle cranial fossa arachnoid cysts (MFACs) can present with symptoms of elevated intracranial pressure (ICP) and/or cyst rupture, warranting further management.

Children with arachnoid cysts may present with headaches, visual symptoms, fatigue, seizures, eye movement abnormalities, or other focal neurological deficits.[1,2,4,6,8-10] Symptom onset is thought to be related to large cyst size and/or cyst rupture, a phenomenon that is characterized by the development of an intracranial hematoma and/or hygroma in the subdural and intracystic spaces.[11-13] Although cyst rupture rates vary, estimates range between 2.3% and 6%.[14-17] The etiology of rupture is often unidentified; however,

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factors such as recent head trauma, including minor trauma that may go unrecognized; cyst size; and cyst location have been associated with an increased risk of rupture.[14,15,18-20] In some instances, cyst rupture may occur spontaneously.[11] Cyst fenestration and drainage of hygroma or hematomas may address symptomatic arachnoid cysts, whether ruptured or not.[1,6,11,21]

Although the neurological presentation of pediatric MFACs is well described, objective measures used to evaluate elevated ICP and guide potential management of these cysts remains a salient gap in the literature. By examining the incidence and factors associated with optic nerve edema and elevated ICP in pediatric MFACs at our institution, we aimed to characterize the clinical utility of ophthalmologic evaluations when treating MFACs.

Methods

This study was approved by the Institutional Review Board at Boston Children's Hospital.

Study population

A clinical research database that includes electronic medical record (EMR) documentation of all neurosurgical, neurological, and ophthalmologic patients at our pediatric institution was retrospectively queried for free-text mention of “arachnoid cyst,” “papilledema,” “intracranial pressure,” and/or “ophthalmology.” Patients who met the criteria for a formal diagnosis of an MFAC as specified by clinician evaluation in their EMR were reviewed. Patients were excluded if they had concurrent tumor diagnoses, were aged greater than 21 years, had fewer than two documented clinical visits, lacked documentation of a formal ophthalmology evaluation, had optic nerve edema diagnosis by a nonophthalmologic clinician (i.e., neurosurgeon, intensive care, or emergency department physician), or were referred to ophthalmology only after surgery for their arachnoid cyst (i.e., they lacked a preoperative baseline evaluation).

Demographic, clinical, and radiographic data were abstracted at the time of initial clinical presentation, first postoperative follow-up, and last documented neurosurgical or neurological follow-up. Clinical information included symptoms of elevated ICP, such as headache, cranial nerve palsy and/or vision abnormalities, emesis, altered mental status, or lethargy, and other potentially related symptoms, such as fatigue, seizures, and a history of head trauma (any, chronic, or acute trauma). The presence of symptoms and history of known head trauma were also used to inform the etiology of rupture (spontaneous or traumatic), with traumatic rupture being classified as such if head trauma and symptom onset aligned or head trauma leading to radiographic evaluation identified rupture shortly after the mechanism of injury. Additional radiographic information included Galassi type, cyst size, presence of rupture, nature of rupture (acute, subacute, chronic), and the type of intracranial fluid collection (hematoma or hygroma) associated with rupture. Galassi type was assigned in accordance with Galassi et al.[22] criteria by authors S.D.R. and A.P.S. and defined as follows: type I is characterized by a small, spindle shape that is limited to the anterior aspect of the middle cranial fossa; type II involves a medium-sized lesion that occupies the anterior and middle temporal fossa with superior extension along the sylvian fissure and

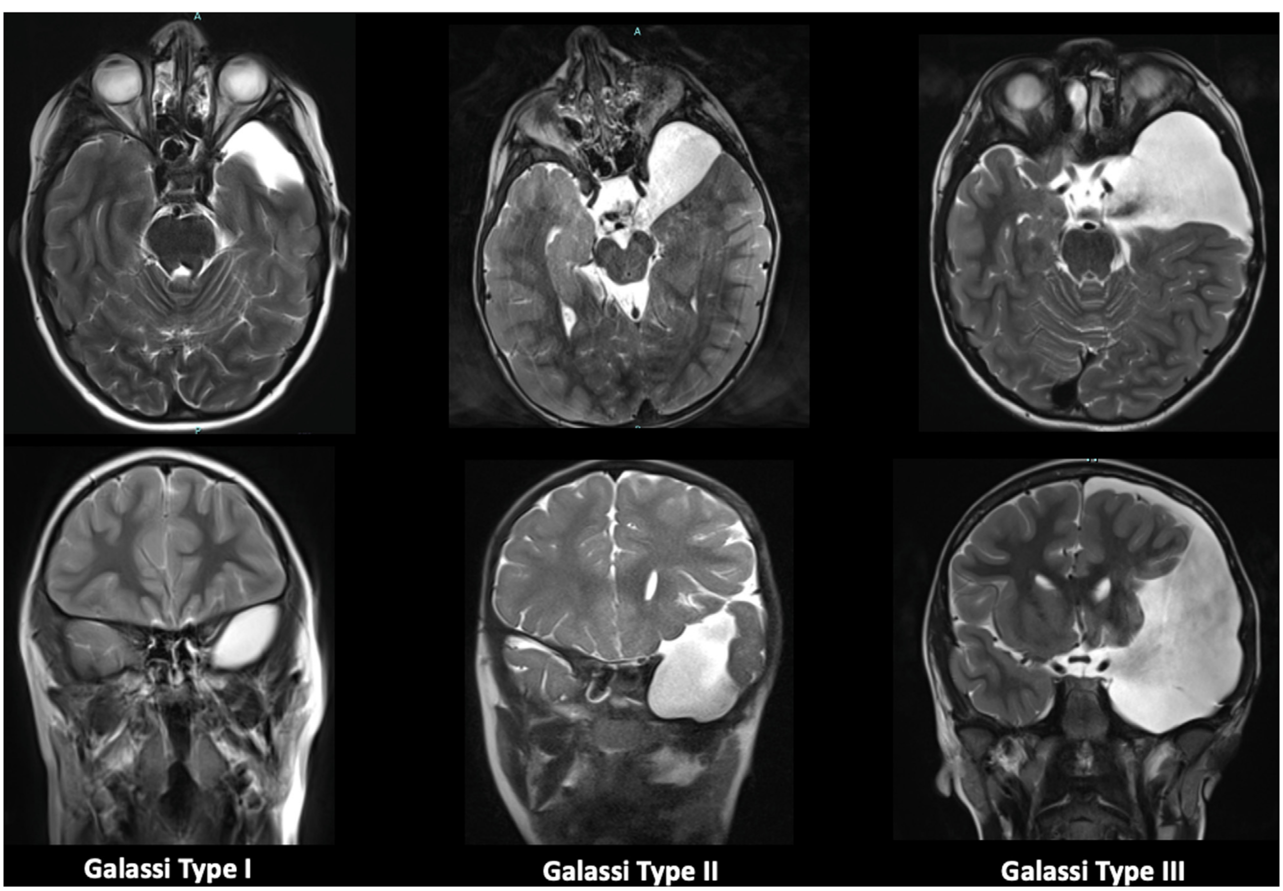

FIGURE 1. Galassi grading system. Type I: Small and limited to the anterior portion of the middle fossa, below the sphenoid ridge. Type II: Extending along the sylvian fissure and can have displacement of the temporal lobe. Type III: Large cyst that encompasses the entire middle fossa with displacement of the temporal, frontal, and/or parietal lobes.

displaces the temporal lobe; and type III is large and occupies the majority of the temporal fossa with displacement of the frontal and parietal lobes (Fig 1). Ophthalmologic information, namely, the presence of disc edema (Fig 2) and/or mention of optic nerve edema, was collected at each patient's preoperative baseline visit and, if the patient was evaluated after surgery, at all postoperative time points.

Statistical analysis

SAS (version 9.4) and R (version 4.2.2) were used for data analysis. General descriptive statistics were applied for categorical and continuous variables. Continuous data were presented as medians and interquartile ranges (IQRs) (quartile 1, quartile 3) due to nonparametric distributions. Categorical data were reported as frequencies and percentages. We performed statistical testing for univariate group comparisons to determine factors associated with optic nerve edema status using the nonparametric Wilcoxon rank-sum test for continuous data and Pearson χ^2 or Fisher exact tests for categorical variables.

To further evaluate potential predictors of optic nerve edema, we conducted a multivariable logistic regression analysis of factors that were statistically significant on univariate analysis, and/or clinically meaningful in our patient population. However, due to our smaller sample size and complete association between optic nerve edema and one of our independent variables, rupture status, we experienced quasi-complete separation of our data. To correct for this separation, we utilized Firth bias-reduced logistic regression model for our multivariable analyses. To further ensure the statistical testing for this separation was appropriate, we also applied an intercept correction to Firth logistic regression for our multivariable analyses.[23] For all models, statistical significance was set at $P < 0.05$. Since there were measurements with a sample proportion of zero, we also applied Jeffrey interval estimate of the binomial proportion confidence interval.

Results

Overall MFAC demographics

We retrospectively reviewed 783 patients from January 1995 to December 2022 who were diagnosed with an MFAC at our institution. Upon further review, between October 2002 and December 2022, we identified 51 patients (6.5%) who met our criteria for having received a formal fundoscopic examination for evaluation of ocular abnormalities and optic nerve edema; these 51 patients were included in the present study.

In our series, there was a male sex and white race predominance (N = 41, 80.4% and N = 37, 72.5%, respectively). At the time of diagnosis, patients were a median of 7.75 years old (IQR: 3.8, 11.7) and the majority of patients presented symptomatically with concern for rising ICP (N = 43, 84.3%) and/or experienced delayed

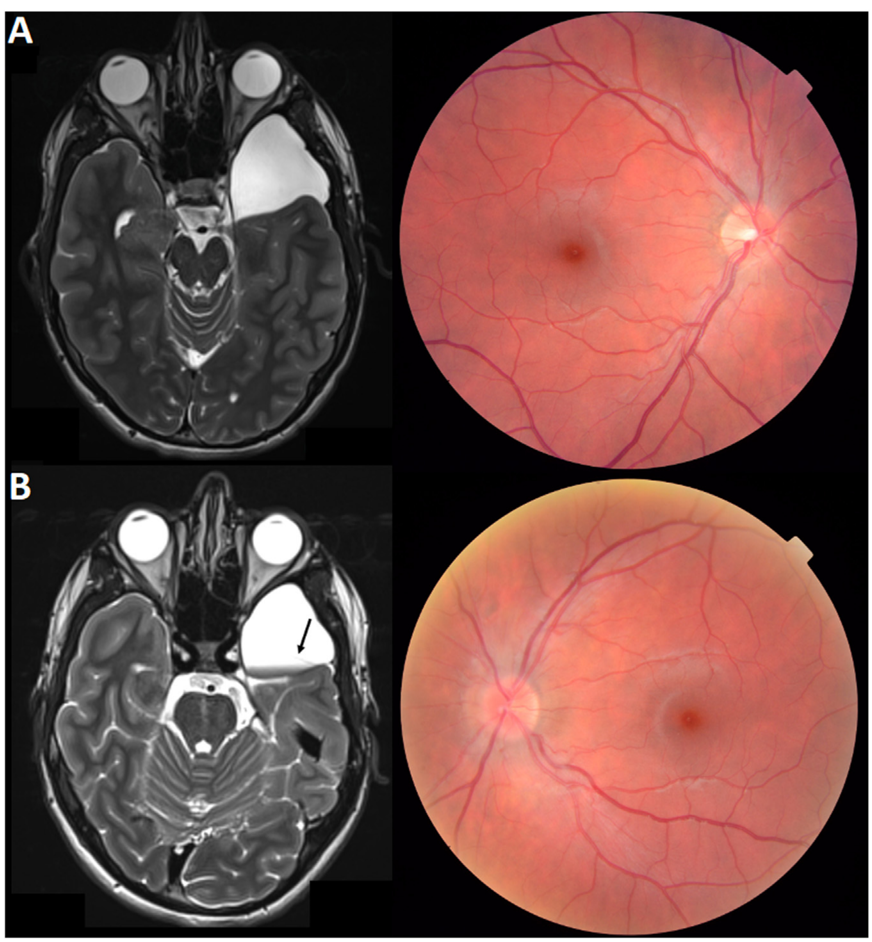

FIGURE 2. Development of optic nerve edema after cyst rupture: (A) Baseline: Axial T2 magnetic resonance imaging (MRI) demonstrating left middle fossa arachnoid cyst, Galassi type I. (B) After rupture: Development of optic nerve edema in left eye only and axial T2 MRI demonstrating hemorrhage (arrow) within ruptured cyst cavity. The color version of this figure is available in the online edition.

symptom onset (N = 6, 11.8%). Headache was the most common symptom (N = 35, 68.6%), which was often characterized by a frequency of daily (N = 9/35, 26%) to weekly (N = 13/35, 37%) headaches, although in some cases the frequency was less severe and noted to be occurring monthly (N = 6/35, 17%) or only occasionally (N = 5/35, 14%); the nature of headache frequency was unavailable for a single patient (N = 1/35, 3%). Other common symptoms included nausea/emesis (N = 20, 39.2%) and vision abnormalities (N = 18, 35.3%). Upon clinical presentation, 19 (37.3%) patients reported a history of previous head trauma, including both acute and chronic trauma. Most patients (n = 37, 72.5%) were referred to ophthalmology by their neurosurgical provider due to concern for elevated ICP in the setting of a concurrent radiographic diagnosis with an MFAC and symptomatic presentation. The remaining patients (n = 14, 37.8%) were referred by neurologists or emergency department physicians because of ongoing symptoms or a history of head trauma in conjunction with a radiographic diagnosis of an MFAC. For further clinical presentation data, refer to Table 1.

Most patients were found to have a single arachnoid cyst (N = 46, 90.2%) on radiographic imaging (Table 2). Among our patients, there was a slightly higher prevalence of left (N = 33, 64.7%) greater than right (N = 13, 25.5%) primary cyst laterality. Cysts were a median volume of 169.9 cm^3 (IQR: 70.5, 647.7 cm^3), and just over half (N = 26/49, 53.1%) were classified as Galassi type II. Radiographic size information was unavailable for two patients who presented with a ruptured cyst and lacked baseline imaging. Evidence of radiographic cyst rupture with the presence of subdural hematoma and/or hygroma was found in 19 (37.3%) total cases, 10 (N = 10/19, 52.6%) of whom had both subdural hygroma and hematoma, five (N = 5/19, 26.3%) with only a subdural hygroma, and four (N = 4/19, 21.1%) with only a subdural hematoma. In our series, spontaneous rupture was a slightly more common occurrence (Table 1), as 11 (N = 11/51, 21.6%) patients were believed to have experienced cyst rupture in the absence of experiencing any acute trauma, whereas eight (N = 8/51, 15.7%) were noted to have ruptured following instances of acute trauma.

Optic nerve edema versus no edema cohort characteristics

Eleven (N = 11/51, 21.6%) patients were found to have optic nerve edema on a formal fundoscopic examination. Patients in this cohort tended to be older at the time of diagnosis with their MFAC than those who did not develop optic nerve edema (median age 10.5 years vs 7.4 years, $P = 0.11$). Most children (N = 8/11, 72.7%) developed bilateral optic nerve edema, although three (27.3%) experienced unilateral optic nerve edema (Fig 2). Four of the 11 patients had grades of optic edema reported in their EMRs, and all four of these patients had similarly-sized, Galassi grade II cysts. One patient was found to have Frisen grade 1 to 2 unilateral disc edema, whereas two other patients had bilateral disc edema grade 2 to 3 and the final patient had bilateral grade 3 to 4 edema. All patients with optic nerve edema were symptomatic; compared with their nonedema counterparts, patients with optic nerve edema presented more often with headache, nausea/emesis, visual symptoms, and lethargy, although these differences were not significant between groups (Table 1). Additionally, the median cyst volume among patients with optic nerve edema tended to be greater than that of patients who did not have any optic nerve edema (218.1 cm^3 vs 112.6 cm^3, $P = 0.20$, Table 2). Interestingly, patients with optic nerve edema had a significantly different distribution of Galassi types compared with the nonedema group, with Galassi type II being more common among the optic nerve edema cohort ($P = 0.016$). However, neither cyst size (diameter, volume) nor elevated ICP symptoms (headache, emesis, vision abnormalities, cranial nerve palsy, altered mental status, and fatigue) was significantly associated with optic nerve edema.

TABLE 1.
Demographic and Clinical Characteristics of 51 Pediatric Patients With Middle Fossa Arachnoid Cysts by Optic Nerve Edema Status

Characteristic	Overall Cohort (n = 51)	Optic Nerve Edema (n = 11)	No Optic Nerve Edema (n = 40)	*P* Value
Age at diagnosis (years)	7.75 (3.8, 11.7)	10.5 (7.4, 13.4)	7.42 (2.5, 11.6)	0.11
Male	41 (80.4%)	8 (72.7%)	33 (82.5%)	0.67
Race				
White	37 (72.5%)	9 (81.8%)	28 (70%)	-
Other	8 (15.7%)	2 (18.2%)	6 (15%)	
Unknown	6 (11.8%)	-	6 (15%)	
Patient history				
History of head trauma	19 (37.3%)	8 (72.7%)	11 (27.5%)	0.012*
Clinical presentation				
Symptomatic	43 (84.3%)	11 (100%)	32 (80%)	0.44
Delayed symptom onset†	6 (11.8%)	-	6 (15%)	
Asymptomatic	2 (3.9%)	-	-	
Headache	35 (68.6%)	10 (90.9%)	25 (62.5%)	0.14
Nausea/emesis	20 (39.2%)	7 (63.6%)	13 (32.5%)	0.085
Agitation	8 (15.7%)	1 (9.1%)	7 (17.5%)	0.67
Motor/gait abnormalities	7 (13.7%)	1 (9.1%)	6 (15%)	1.0
Sensory changes	2 (3.9%)	-	2 (5%)	1.0
Visual changes	18 (35.3%)	6 (54.5%)	12 (30%)	0.25
Macrocephaly	4 (7.8%)	-	4 (10%)	0.57
Seizures	7 (13.7%)	-	7 (17.5%)	0.32
Lethargy	3 (5.9%)	1 (9.1%)	2 (5%)	0.53
Rupture (any)	19 (37.3%)	11 (100%)	8 (20%)	<0.0001*
Spontaneous	11 (21.6%)	5 (45.5%)	6 (15%)	0.078
Traumatic	8 (15.7%)	6 (54.5%)	2 (5%)	0.0006*
Years from presentation to last follow-up‡	1.6 (0.6, 5.0)	2.4 (0.2, 3.5)	1.6 (0.65, 5.6)	0.46
Lost to follow-up	2 (3.9%)	1 (9.1%)	1 (2.5%)	1.0

Categorical variables are presented as n (%) and continuous variables as median (interquartile range).
* Indicates statistical significance with $P < 0.05$.
† Six patients were asymptomatic on presentation to our institution, but they later became symptomatic within two years of this initial presentation.
‡ Measured from clinical presentation to last neurological (neurology or neurosurgery) or ophthalmologic clinic visit.

TABLE 2.
Radiographic Characteristics of 51 Pediatric Patients With Middle Fossa Arachnoid Cysts by Optic Nerve Edema Status

Characteristic	Overall Cohort (n = 51)	Optic Nerve Edema (n = 11)	No Optic Nerve Edema (n = 40)	*P* Value
Laterality				
Unilateral MFAC	46 (90.2%)	10 (90.9%)	36 (90%)	1.0
Bilateral MFAC	5 (9.8%)	1 (9.1%)	4 (10%)	
Left laterality	33 (64.7%)	6 (54.5%)	27 (67.5%)	0.59
Right laterality	13 (25.5%)	4 (36.4%)	9 (22.5%)	
Cyst maximum diameter (mm)*	42 (35, 60)	43 (39.3,46.3)	40 (33, 62.5)	0.63
Volume (cm^3)*	169.9 (70.5, 647.7)	218.1 (173.2, 285.0)	112.6 (60.8, 712.9)	0.20
Galassi grade*				
Galassi I	14 (28.6%)	-	14 (35.9%)	0.016†
Galassi II	26 (53.1%)	9 (90%)	17 (43.6%)	
Galassi III	9 (18.4%)	1 (10%)	8 (20.5%)	
Intracranial bleed				
Subdural hygroma	15 (29.4%)	9 (81.8%)	6 (40.0%)	<0.0001†
Subdural hematoma	14 (27.5%)	9 (81.8%)	5 (12.5%)	<0.0001†
Radiographic rerupture of cyst	2 (3.9%)	1 (9.1%)	1 (2.5%)	1.0
Hydrocephalus	2 (3.9%)	-	2 (5%)	1.0

Abbreviation:
MFAC = Middle fossa arachnoid cyst
Categorical variables are presented as n (%) and continuous variables as median (interquartile range).
* Measurements were missing for two patients who lacked preoperative baseline imaging and presented to our institution upon cyst rupture.
† Indicates statistical significance with $P < 0.05$.

None of the patients with unruptured MFAC had optic nerve edema (95% interval of 0% to 7%); all patients in the optic nerve edema cohort had a ruptured arachnoid cyst, with traumatic cyst rupture being slightly more common than spontaneous rupture (N = 6/11, 54.5% vs N = 5/11, 45.5%, respectively). Conversely, among the patients without edema, cyst rupture was less frequent (N = 8/41, 19.5%) and was more often spontaneous than related to acute head trauma (Table 1). On univariate analysis, cyst rupture was significantly associated with optic nerve edema status ($P < 0.0001$, Tables 1 and 2). In particular, patients who experienced traumatic cyst rupture were significantly more likely to develop optic nerve edema than patients who experienced spontaneous rupture ($P = 0.0006$, Table 1). On multivariable analysis of cyst size (Galassi type), symptomatic presentation, rupture status, and traumatic cyst rupture, the two variables rupture status and optic nerve edema were initially found to have quasi-complete separation. After correcting for this using a Firth bias-reduced logistic regression model, only rupture status retained strong statistical significance with optic nerve edema ($P = 0.0078$, Table 3).

Patients with optic nerve edema (N = 10, 90.9%) were also significantly more likely to undergo neurosurgical intervention than patients without ($P < 0.001$, Table 4), with surgery occurring a median of 2 (IQR: 1.0, 6) days following their initial ophthalmologic evaluation and a median of 7.5 (IQR: 3.25, 40) days following initial presentation to our institution. The majority of patients underwent microsurgical fenestration of their cyst into the adjacent subarachnoid spaces and basal cisterns (N = 9/10, 90%), sometimes in conjunction with evacuation of a subdural collection, whereas a single patient only received evacuation of subdural bleed (N = 1/10, 10%) (Table 4). Before cyst fenestration, a single patient had a lumbar puncture (LP) and associated pressure (13 cm H_2O) documented, although this was unavailable for the majority of operative patients in our series.

TABLE 3.
Multivariate Analysis of Factors Associated With Optic Nerve Edema Status

Covariate	Optic Nerve Edema Status	
	Adjusted Coefficient (95% CI)	*P* Value
Symptomatic Presentation		
Symptomatic	−2.1 (−7.5, 3.3)	0.37
Delayed symptom onset	−0.73 (−6.1, 4.7)	0.74
Galassi grade		
Grade 2	0.73 (−2.2, 5.6)	0.21
Grade 3	0.92 (−2, 6.0)	0.64
Cyst rupture	3.1 (0.72, 8.0)	0.0078*
Traumatic cyst rupture	1.8 (−0.33, 4.4)	0.10

Abbreviation:
CI = Confidence interval
* Indicates statistical significance with $P < 0.05$.

After surgery, there was an association between optic nerve edema and the development of postoperative complications compared with patients who received an operative intervention in the nonedema cohort (N = 2 vs N = 0 complications, $P = 0.037$). A single patient with optic nerve edema developed a complication of bilateral subdural hematomas during the hospital stay, resulting in a return to the operating room 11 days after the cyst fenestration surgery for placement of a cystoperitoneal shunt. The patient required a subsequent procedure three months later for drainage of chronic subdural collection via the placement of a subdural drain. The other patient who experienced a postoperative complication had previously undergone surgery to evacuate the subdural hematoma followed by staged cyst fenestration six weeks later following cyst rerupture. However, status post this subsequent fenestration, there was evidence of interval cyst hemorrhage and an epidural fluid collection containing blood, cerebrospinal fluid, and air underlying the craniotomy, which resolved with conservative management over the following postoperative month.

Following cyst fenestration and evacuation of rupture-associated subdural collections, formal ophthalmologic documentation of postoperative optic nerve edema improvement (N = 2/8, 25%) and/or complete resolution (N = 6/8, 75%) was noted in eight of the 10 surgical cases (80%). The time to resolution or improvement in optic nerve edema was a median of 5 and 1.5 postoperative months, respectively. Given that not all patients had longitudinal ophthalmologic follow-up, we remained conservative in our classification of those who experienced improvement, but not complete resolution, of their optic nerve edema and classified these patients as such if their postoperative fundoscopic examination demonstrated that their optic nerve edema was still present but improving. Postoperative ophthalmologic evaluation was not available for the remaining two operative patients. At time of last follow-up, a median 1.8 (IQR: 0.4, 2.9) years after surgery, half (N = 5/10, 50%) of the patients who received surgery for their optic nerve edema endorsed complete symptom resolution, whereas the

TABLE 4.
Management and Outcome Characteristics of 51 Pediatric Patients With Middle Fossa Arachnoid Cysts by Optic Nerve Edema Status

Characteristic	Overall Cohort (n = 51)	Optic Nerve Edema (n = 11)	No Optic Nerve Edema (n = 40)	*P* Value
Management				
Surgery	18 (35.3%)	10 (90.9%)	8 (20%)	<0.0001*
Observation	33 (64.7%)	1 (9.1%)	32 (80%)	
Surgery type				
Cyst fenestration	14 (27.5%)	7 (63.6%)	7 (17.5%)	1.0
Evacuation/drainage of bleed	1 (2%)	1 (9.1%)	-	
Both	3 (5.9%)	2 (18.2%)	1 (2.5%)	
Age at surgery (years)	8.9 (6.8, 14.1)	10.9 (8.0, 15.3)	7.4 (4.5, 8.6)	0.10
Time from ophthalmologic evaluation to surgery (days)	3 (1, 37)	2 (1, 6)	43 (1, 144.5)	0.24
Time from clinical presentation to surgery (days)	18.5 (1.5, 79)	7.5 (3.25, 40)	69 (0.8, 1093)	0.42
Perioperative complications	2 (3.9%)	2 (18.2%)	-	0.037*
Need for cerebrospinal fluid diversion procedure	1 (2%)	-	1.00	1.0
Time to last follow-up (years)	2.20 (0.67, 6.0)	3.02 (0.86, 11.2)	2.11 (0.69, 6.0)	0.80
Last follow-up outcomes				
Lost to follow-up	2 (4.1%)	-	2 (5.3%)	1.0
Symptomatic	28 (54.9%)	6 (54.5%)	22 (55%)	0.19
Papilledema resolution	8 (15.7%)	8 (72.7%)	-	1.0
Headache	18 (35.2%)	5 (45.5%)	13 (32.5%)	0.73
Nausea	3 (6.1%)	-	3 (7.9%)	1.0
Motor/gait abnormalities	1 (2%)	-	1 (2.6%)	1.0
Macrocephaly	1 (2%)	-	1 (2.6%)	1.0
Seizures	7 (14.3%)	-	7 (18.4%)	1.0
Change in symptoms				
Symptom resolution	20 (39.2%)	5 (45.5%)	15 (37.5%)	0.023*
Symptom improvement	12 (24.5%)	6 (54.5%)	6 (15.8%)	
Symptom worsening	2 (4.1%)	-	2 (5.3%)	
No change	15 (29.4)	-	15 (37.5%)	

Categorical variables are presented as n (%) and continuous variables as median (interquartile range).
* Indicates statistical significance with $P < 0.05$.

other half (N = 5/10, 50%) reported overall symptom improvement as their visual symptoms, lethargy, motor weakness, agitation, and emesis resolved; however, they continued to experience occasional headaches (Table 3). A single patient did not undergo surgery due to spontaneous resolution of the optic nerve edema and improvement of the sixth nerve palsy 75 days after the initial ophthalmologic evaluation while under ophthalmologic surveillance due to parental aversion to surgery.

Patients without edema were less likely to undergo neurosurgical intervention compared with the optic nerve edema cohort (Table 3, Supplemental Table 1). However, the patients without edema who did go on to receive a surgical intervention (N = 8), tended to receive surgery later than their optic nerve edema counterparts, at a median of 43 days (IQR: 1, 144.5) following their initial ophthalmologic evaluation and 69 days (IQR: 0.75, 1093) since presentation to our institution ($P = 0.24$ and $P = 0.42$, respectively). Among these eight operative patients without papilledema, cyst rupture was not associated with surgical status ($P = 0.65$), as only two experienced cyst rupture (Supplemental Table 1). For the remaining six operative patients, the indications for surgery included symptom worsening and/or enlarging cyst size. Similar to the surgical approach in our optic nerve edema cohort, all operative patients without edema underwent microsurgical fenestration of their respective arachnoid cysts.

Among the nonoperative patients without edema (N = 32), there were six patients who experienced cyst rupture who did not receive surgery, and instead, were conservatively managed with subsequent neurosurgical and neurological follow-up. A small subset of these nonoperative patients, two of whom had cyst rupture, also received LPs due to concern for elevated ICP. The opening pressures for these patients were 25, 29, and 36 cm H_2O, respectively, although the pressure for the fourth patient was not documented.

At the time of last follow-up, four (N = 4/8, 50%) nonedema patients reported complete symptom resolution following surgery for their MFACs. Three (n = 3/8, 37.5%) operative but nonedema patients reported symptom improvement in their nausea/emesis, visual changes, and the severity of their headaches, although they continued to experience occasional mild headaches. A single operative, nonedema patient experienced no change in their headache symptoms after surgery. Among the nonoperative patients in this nonedema cohort, 14 (N = 14/30, 46.7%) reported no change in their headaches, nausea/emesis, seizure activity, motor deficits, and/or mood disturbances. Eleven nonoperative patients (N = 11/30, 36.6%) reported symptom resolution, and three (N = 3/30, 10%) reported symptom improvement with regard to headache severity and frequency. Two nonoperative patients (N = 2/30, 6.7%) endorsed symptom worsening, characterized by new onset of seizures, at the time of last follow-up. Two patients were lost to follow-up, and two patients remained asymptomatic.

Discussion

The middle cranial fossa is a common location for arachnoid cysts in children.[1,2,5-7] Although the vast majority of MFACs are benign entities discovered incidentally on neuroimaging,[1-4] in a subset of patients, cysts may cause elevated ICP. The utility of specialized fundoscopic examination for evaluating rising ICP in patients with MFACs is not well described, especially in a patient population involving both ruptured and unruptured cysts. The present study reviewed all pediatric patients diagnosed with MFACs at our institution who were referred for a formal ophthalmologic examination.

Although the natural history and neurological manifestations of arachnoid cysts are well reported, the clinical utility of ophthalmologic evaluation during MFAC management remained largely unexamined. All 51 children in our study were referred to ophthalmology before any surgical intervention. Although other providers also evaluate for optic nerve edema, we sought to improve the homogeneity of the data by relying on highly expert evaluation. Upon examination, 21.6% of the patients with MFACs had optic nerve edema. This is one of the largest series to report on the presence of optic nerve edema in pediatric patients with MFACs who have been evaluated by ophthalmology. In line with other series,[24-27] most (72.7%) of our patients developed bilateral optic nerve edema in the setting of rising ICP, although unilateral optic nerve edema was found in select cases (27.3%).[28] The exact cause of unilateral optic nerve edema remains unknown due to its rare occurrence; however, some theories suggest that optic nerve sheath abnormalities and/or a lack of communication between the orbit and subarachnoid space may be related.[28-31]

Prior series have reported a male sex predominance and left cyst laterality among patients diagnosed with intracranial arachnoid cysts.[4,32,33] Similarly, in our study of pediatric MFACs, most patients were male and had left lateralized cysts. That being said, this cohort is enriched for symptomatic presentation, which is uncommon in the overall population of MFACs.[1,5,34] Selection bias is a common limitation of retrospective clinical series and may explain this phenomenon, but it may also limit the generalizability of our findings. This bias could involve referral patterns for evaluation by an ophthalmologist, such as symptomatology or radiographic features that increase suspicion for elevated ICP. The overall management of this cohort was largely driven by the presence of symptoms, and patients found to have optic nerve edema were significantly more likely to undergo neurosurgical intervention than those without. Surgery moderately improved the natural history of symptoms both in patients with and without edema to a similar degree. Comparatively, there were lower rates of improvement and a slightly higher tendency for symptom worsening in the nonoperative cohort. Thus, we propose that these data derive from a management paradigm representing a reasonable referral pattern of symptomatic MFAC for specialist ophthalmology evaluation of optic nerve edema.

In this cohort, optic nerve edema was not observed in the absence of other neurological symptoms. Compared with the nonedema cohort, patients with optic nerve edema most commonly presented with headaches or vision abnormalities. Furthermore, optic nerve edema was always observed in the context of cyst rupture, characterized by radiographic evidence of an associated subdural fluid collection. Prior studies show that large cyst volume and recent head trauma are associated with a higher risk of rupture, particularly for cysts with >5 cm maximal diameter[11,14] and that rupture is often correlated with symptomatic presentation.[14,15,18,19] For this reason, even though the relationship between optic nerve edema, cyst size, and symptoms of elevated ICP has not been well described, we hypothesized that these factors could be predictors of optic nerve edema. We observed that patients who experienced cyst rupture, in particular rupture following an instance of acute trauma, were more likely to present with optic nerve edema ($P < 0.0001$). However, when accounting for the rupture status (ruptured versus unruptured), rupture type (traumatic versus spontaneous), cyst size (Galassi type), and symptomatic presentation on multivariable analysis, only cyst rupture status remained significantly associated with optic nerve edema in our series. This association may be due, in part, to the frequently congenital nature of arachnoid cysts, as the size of the cyst may be well accommodated except in the setting of acute volume change during a rupture. Alternatively, we did not observe optic nerve edema in patients who did not have cyst rupture.

Among the patients who experienced cyst rupture in our series, 57.9% were found to have optic nerve edema. This finding aligns well with a recent study by Mayeda et al.,[24] which reports similarly high rates of papilledema in pediatric patients with ruptured arachnoid cysts, although these cysts were not limited to the middle cranial fossa. In providing a comparison between the ophthalmologic evaluations of ruptured and unruptured MFACs, our series complements the findings by Mayeda et al.[24] and further underscores the significance of referring patients with ruptured arachnoid cysts for a formal fundoscopic examination.

As mentioned, patients with optic nerve edema were significantly more likely to undergo neurosurgical intervention than clinical observation compared with those without ($P < 0.0001$). In our series, 10 of the 11 (91%) patients with optic nerve edema received surgery. These findings further validate a recent provider survey, which assessed decision-making trends when managing intracranial arachnoid cysts.[35] There was a uniform improvement in symptoms among these patients who received surgery for their ruptured cyst, which aligns with prior reports of favorable clinical outcomes following the surgical management of intracranial arachnoid cysts.[1,6,7,11,36,37] However, in the absence of optic nerve edema, cyst rupture was not associated with surgical status, as demonstrated by the low rate of cyst rupture in patients without edema who received cyst fenestration. Thus, it is important to note that in the absence of elevated ICP and optic nerve edema, surgery may not be warranted and can increase the risk of additional complications,[38] the latter of which was experienced by two operative patients in our series. Although this translated to an overall low rate of postoperative complications in our study, patients who receive surgery for their MFAC, especially through cyst fenestration as was common in our series, are at risk for perioperative complications like the development of new subdural collections or hematomas, CSF leaks, wound dehiscence, and infection, among others.[38-40]

Despite these risks, among the eight operative patients who had postoperative ophthalmologic evaluations, all experienced resolution or improvement in their optic nerve edema at a median of 5 and 1.5 postoperative months, respectively. Symptom improvement, characterized by complete symptom resolution in one case and improvement in all symptoms except for mild headaches in the other, was reported for the remaining two operative patients with optic nerve edema. Unfortunately, follow-up ophthalmologic information was unavailable for these two patients, and this was only determined clinically at their neurosurgical follow-up. As a result, these findings suggest that ophthalmologic consultation for optic nerve edema in patients with ruptured cysts may be a helpful metric for determining whether neurosurgical intervention is indicated. However, further study into other factors influencing the neurosurgical management of MFACs is warranted.

A single patient with optic nerve edema did not receive surgery in our series. This patient suffered minor head trauma before clinical presentation and presented with a left ruptured MFAC. Surgery was recommended as the mode of most likely and most rapid improvement by our neurosurgical team; however, parental aversion to surgery led to a strategy of close clinical observation for two months, during which the symptoms and optic nerve edema gradually improved. Although it is uncommon for optic nerve edema to resolve spontaneously, Deveney et al.[25] report a similar case, where a symptomatic patient with a left-lateralized, ruptured MFAC underwent ophthalmologic surveillance and was found to have preoperative resolution of their symptoms and papilledema within a week and month, respectively, of initial presentation. This phenomenon was further observed in a small series by Maher et al.,[26] where a patient with a ruptured MFAC associated with a subdural hygroma, bilateral papilledema, and left cranial nerve VI palsy experienced markedly improved symptoms two days following presentation, leading to the cancellation of a planned surgical intervention. Although the spontaneous resolution of optic nerve edema is rare in MFACs, these cases further illustrate the importance of ophthalmologic monitoring for guiding cyst management. These findings also highlight the potential benefit of referring patients with optic nerve edema to ophthalmologic surveillance should spontaneous symptom improvement occur, or the indication for surgery become less clear.

Because symptoms at presentation of intracranial arachnoid cysts are often nonspecific,[4,25,41] optic nerve edema may be a valuable sign of elevated ICP that may help guide management and the need for operative intervention. In particular, pediatric patients with MFACs who experienced recent, acute head trauma and cyst rupture may benefit the most from formal ophthalmologic evaluation; however, this evaluation may not be as valuable for patients lacking a history of head trauma or who present asymptomatically without cyst rupture.

Limitations

Given our study's small sample size and retrospective, single-institution design, our findings may not be generalizable to a more diverse population of pediatric MFACs. As such, our study has the limitations inherent to a retrospective study such as what data are available, like the details and nuances of clinical history and specialist evaluation. In particular, our cohort contains selection bias because we do not universally refer all patients with MFACs for an ophthalmology evaluation at our institution, thus our sample does not comprehensively evaluate the true incidence of optic nerve edema and rising ICP in patients with asymptomatic and unruptured MFACs. In this context, assuming that patients with unruptured MFACs referred for ophthalmology evaluation are similar to those who were not referred for ophthalmology, there may be no risk of optic nerve edema in patients with unruptured MFACs, with a 95% confidence interval of 0% to 7%. Nonetheless, a larger sample size, perhaps through a multi-institution design, is needed to increase the overall statistical power and applicability of our findings to a broader pediatric MFAC population.

Our study was further limited by the fact that many of our included patients did not receive an LP as we do not routinely perform LPs for patients with MFACs at our institution. As a result, our study lacks commentary on the potential utility of additional objective measures for elevated ICP. Owing to the young age of our patients and the need for parent/guardian input at clinical presentation and follow-up visits, recall bias may have also occurred when patient symptoms and trauma history were recorded in the EMR. Finally, our study only evaluated pediatric arachnoid cysts located in the middle cranial fossa, yet MFACs are rare in the general population and characterize only one subtype of intracranial arachnoid cyst. Thus, further investigation into the incidence of optic nerve edema and factors underlying elevated ICP in a much broader range of intracranial arachnoid cyst types is warranted.

Conclusions

In our series of 51 patients with pediatric MFACs that had been evaluated by ophthalmology, optic nerve edema occurred in 21.6% of cases. Among these patients, cyst rupture was always associated with optic nerve edema, whereas we did not observe optic nerve edema in patients with unruptured cysts. Among all the patients ($n = 19$) who experienced cyst rupture in our series, over half (57.9%) had optic nerve edema. Neurosurgical intervention involving cyst fenestration was significantly more common in patients who developed optic nerve edema, and after surgery, most patients experienced improvements in their optic nerve edema and overall symptoms. In conjunction with clinical history and intracranial neuroimaging, ophthalmologic evaluation for optic nerve edema may be a useful tool to help guide MFAC management and the indication for operative intervention, particularly in patients who experienced cyst rupture.

CRediT authorship contribution statement

Joanna E. Papadakis: Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Anna L. Slingerland:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Shivani D. Rangwala:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Mark R. Proctor:** Writing – review & editing, Supervision, Resources, Investigation, Conceptualization. **Ankoor S. Shah:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Alfred P. See:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

There are no disclosures or conflicts of interest to report for any of the authors on this manuscript. No study sponsor(s) were involved in study design; the collection, analysis, or interpretation of data; the writing of the report; or the decision to submit the manuscript for publication.

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Supplementary Data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.pediatrneurol.2024.04.008.

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